```
echo 'export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64' >>
~/.bashrc
```

Step 3: Install Docker and allow your user to run it

```
curl -fsSL https://get.docker.com | sh
sudo usermod -aG docker $USER
newgrp docker
docker run hello-world
```

Step 4: Set the hub's listen address

Replace the example address with your machine's LAN IP, which you can find with *hostname -I*.

```
export HZN_LISTEN_IP=192.168.1.50
```

Installing Open Horizon

One script installs the whole all-in-one environment: the management hub services, the agent and the *hzn* CLI. Run it as *root* with your environment variables preserved:

```
sudo -E bash -c "curl -sSL https://raw.githubusercontent.
com/open-horizon/devops/master/mgmt-hub/deploy-mgmt-hub.sh |
bash"
```

The script takes several minutes. It downloads the hub images, starts the Exchange, agbot, CSS, PostgreSQL and MongoDB containers, creates an organisation called *myorg* with an *admin* user, publishes the example services, installs the agent and registers the machine as node *node1*. It ends by printing two credentials that you must save:

```
export HZN_ORG_ID=myorg
export HZN_EXCHANGE_USER_AUTH=admin:<generated-password>
```

Useful script options are:

Flag	Effect
-A	Install the hub and CLI only, without the local agent
-E	Skip loading the example services, policies and patterns
-S	Stop the hub services and agent
-s	Start them again without reinstalling
-P	Purge: delete volumes and images and uninstall the agent

***Installing an agent on another device*:** On a separate edge device such as a Raspberry Pi, point the agent at your hub and run *agent-install.sh:*

```
export HZN_ORG_ID=myorg
export HZN_EXCHANGE_USER_AUTH=admin:<password>
export HZN_EXCHANGE_URL=http://192.168.1.50:3090/v1
export HZN_FSS_CSSURL=http://192.168.1.50:9443/
export HZN_AGBOT_URL=http://192.168.1.50:3111
export HZN_DEVICE_ID=pi-node-01
export HZN_DEVICE_TOKEN=<any-random-string>

curl -sSL https://github.com/open-horizon/anax/releases/
latest/download/agent-install.sh \
  | sudo -s -E bash -s -- -i css: -p IBM/pattern-ibm.
helloworld -w '*' -T 120
```

This installs the agent from the hub's CSS and registers the device with the sample 'hello-world' pattern.

Setting up Open Horizon

Set up means making your credentials permanent and confirming that the hub, the agent and the CLI can talk to each other.

Step 1 – Save the credentials: Add the two lines printed by the installer to *~/.bashrc* so every new terminal has them:

```
echo 'export HZN_ORG_ID=myorg' >> ~/.bashrc
echo 'export HZN_EXCHANGE_USER_AUTH=admin:<generated-
password>' >> ~/.bashrc
source ~/.bashrc
```

Step 2 – Check the installation: Each command below should return without errors.

Command	Expected result
hzn version	Matching CLI and agent versions
hzn exchange status	Exchange version and health details in JSON
hzn exchange user list	The *admin* user of *myorg*
hzn node list	This node, with *configstate = configured*
hzn exchange service list IBM/	The example services published by the installer
docker ps	Running hub containers (exchange, agbot, css, postgres, mongo) and the 'hello-world' service

Step 3 – Create your own user (optional): Working as an admin is fine for learning. In a team, create a named user and an organisation for each tenant:

```
hzn exchange user create -o myorg -A student1 <password>
student1@example.com
hzn exchange user list -o myorg student1
```